When Transport Stops Being a Tautology

Invariants, Non-Invariants, and New Formulas in Twin Arithmetic

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Abstract

The twin construction $x \odot_{eg} y = g^{-1} \cdot \big((g \cdot x) \odot_e (g \cdot y)\big)$ is, for a single operation, a relabeling: by the equivariance principle every twin space is isomorphic to the base, so it produces no new algebraic theorem. Yet, exactly as the Pythagorean theorem becomes the analysis of variance and the convolution theorem becomes the fast Fourier transform, the construction yields major results once it is read in a frame carrying extra structure. We make this precise. We state a *boundary* principle — a property transports iff it lives in the signature the isomorphism preserves; *non-invariant* structure (metric, measure, order, a second operation) is where new tools arise. We show that the coherent bijections fall into exactly *four productive frames* (the homomorphisms between $+$ and $\times$), each tied to an application: linear (rigid), $+ \to \times$ (concentration), $\times \to +$ (entropy), $\times \to \times$ (primes). We then mint concrete results: quasi-arithmetic means and the power-mean inequality from the invariants; *φ-concentration bounds* with tail-adaptive base selection from the metric non-invariant; and, from the multiplicative frame, the identification of affine twin arithmetic with the multiplicative monoid of an arithmetic progression (a Hilbert monoid), with a corrected unit, twisted primes $\{n : 2n+1 \text{ prime}\}$, non-unique factorization, and a measurable Euler-product defect. Every claim is computed and tested (226 passing tests).

Keywords: transport of structure, quasi-arithmetic means, concentration inequalities, primes in arithmetic progressions, Hilbert monoids, non-Newtonian calculus.

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1 Introduction

The companion papers established the twin operation space $\mathrm{Op}(E, \odot_e; G)$ and its group-theoretic discrimination, and the toolkit that turns the choice of a twin domain into a measured decision. Both rest on a candid limitation: by the *equivariance principle* the map $\Psi_g : x \mapsto g^{-1} \cdot x$ is an isomorphism $(E, \odot_e) \xrightarrow{\sim} (E, \odot_{eg})$, so every twin operation is, up to relabeling, the base. For a *single* operation, the construction is a tautology.

This paper argues — and computes — that the tautology is *bounded*, and that the productive zone is large and creative. The guiding analogy is classical: the Pythagorean theorem is one statement about an inner-product space, yet it *is* the orthogonality of uncorrelated random variables, the variance decomposition, and the analysis of variance; the convolution theorem $\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$ is one intertwining identity, yet it *is*, with the fast transform, a major computational method. A relabeling becomes mathematics when it is read in a frame that carries structure the relabeling does *not* preserve.

Contributions. (i) a boundary principle separating invariant from non-invariant properties (§2); (ii) the classification of coherent bijections into four productive frames, each an application

(§3); (iii) new formulas from invariants — quasi-arithmetic means and the power-mean inequality (§4); (iv) new tools from non-invariants — φ-concentration bounds with tail-adaptive base selection, and the multiplicative frame's Hilbert monoids, corrected primes, and Euler-product defect (§5). A by-product is a correction to the literature's twisted-prime unit.

2 The boundary: invariant versus non-invariant

Let $\Psi_g : (E, \odot_e) \to (E, \odot_{eg})$ be the equivariance isomorphism. A property of the structure transports faithfully along Ψ_g precisely when it is expressible in the signature Ψ_g preserves.

Proposition 1 (Boundary principle)**.** *A property P is* invariant *(carried by every Ψ_g) iff it is definable in the reduct $\{\odot\}$ that Ψ_g preserves. A property is* non-invariant *iff it requires part of the signature Ψ_g does not preserve: a metric d, a measure μ, an order $\leq$, a topology, or a second operation $\star$. New tools come exactly from the non-preserved signature.*

Operationally, one reads off what a bijection preserves (Table 1): the identity preserves everything; an affine map is a similarity (order and metric up to scale) but not an additive automorphism; exp is the additive-to-multiplicative homomorphism yet distorts the metric; log preserves only order. The algebraic kernel records the preserved *algebra*; whether Ψ_g is an isometry / measure-preserving / monotone records the preserved *analysis*.

Table 1: What a bijection preserves on a sample (computed by `invariance_report`). "$+\!\to\!\times$" is the homomorphism $\varphi(x{+}y) = \varphi(x)\varphi(y)$.

φ	order	$+\!\to\!+$	$+\!\to\!\times$	isometry	metric distortion
identity	yes	yes	no	yes	$\times$1.0
$2x{+}1$	yes	no	no	yes (scale)	$\times$1.0
exp	yes	no	**yes**	no	$\times$7.4
x^2	yes	no	no	no	$\times$3.7
log	yes	no	no	no	$\times$3.8

3 The four productive frames

A coherent bijection is one realizing a homomorphism between the two operations $+$ and $\times$. There are exactly four, and a sweep of a bijection catalogue (`discover_homomorphism_types`) recovers them automatically (Figure 1, Table 2); everything else is *generic*.

Table 2: The four frames and their applications.

homomorphism	representative φ	application
$+ \to +$	linear $2x$	rigid (addition preserved)
$+ \to \times$	exp	concentration (MGF, Chernoff)
$\times \to +$	log	entropy, log-domain computation
$\times \to \times$	x^p, $1/x$, $\sqrt{\cdot}$	multiplication, primes, zeta
generic	$2x{+}1$, $x^2{+}x$	new arithmetic

Reading. The classification is sharp — each coherent bijection falls in exactly one frame — while the bar heights show the orthogonal, non-invariant axis (metric distortion) that the homomorphism type does not fix. The two together are the whole story: an invariant skeleton, a non-invariant flesh.

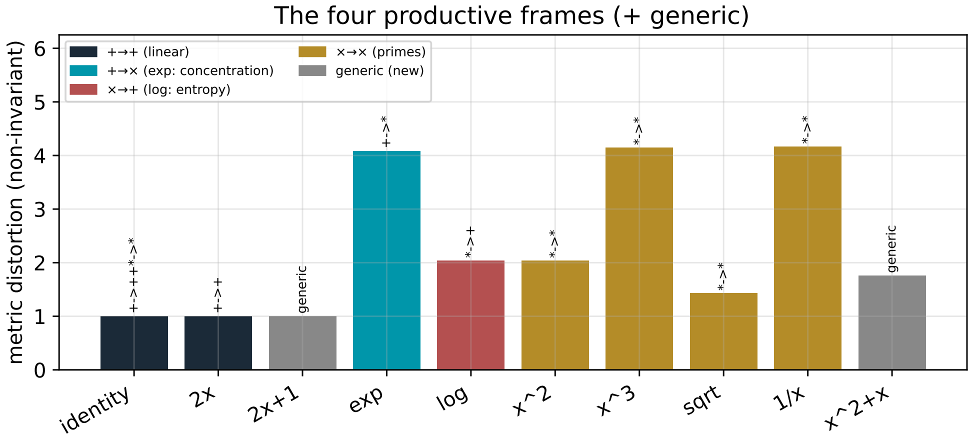

Figure 1: The bijection catalogue classified by homomorphism frame; bar height is the (non-invariant) metric distortion. The four coherent frames colour-coded; generic maps in grey.

Remark 2 (Why exactly these four). The homomorphisms relate the additive and multiplicative structures of the line. The frame $+ \to \times$ is uniquely realised by exp (up to scaling), which is why it, and not another convex map, underlies the moment generating function — a fact we use in §5.

4 New formulas from invariants

Writing an invariant identity in different frames yields a family of concrete formulas. The cleanest example is the *quasi-arithmetic* (Kolmogorov–Nagumo) mean [1, 2]

$$M_\varphi(x) = \varphi^{-1}\Big(\tfrac{1}{n}\textstyle\sum_i \varphi(x_i)\Big),$$

the twin average. For $\varphi = x^p$ it is the power mean M_p; $p = 1$ arithmetic, $p \to 0$ geometric (the log frame), $p = -1$ harmonic.

Proposition 3 (Power-mean inequality). *For $p \le q$, $M_p(x) \le M_q(x)$, with $\min x = M_{-\infty} \le M_p \le M_{+\infty} = \max x$.*

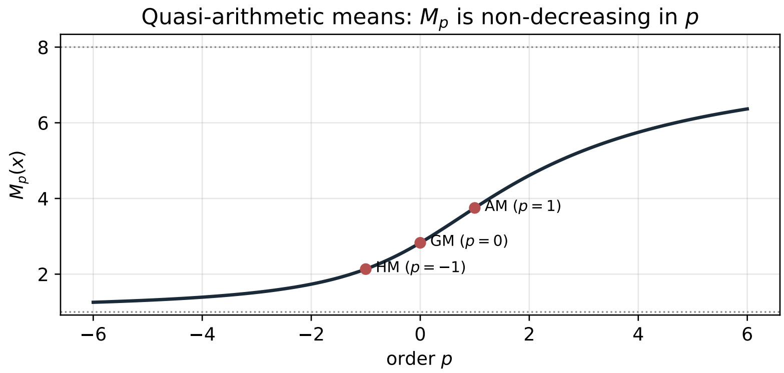

Figure 2: The quasi-arithmetic mean M_p as a function of the order p (one data set), with the harmonic, geometric and arithmetic means marked; M_p is non-decreasing and confined between min and max.

Reading. Each value M_p is, in isolation, "the same" average viewed in the frame $\varphi = x^p$ — an invariant. The *monotonicity across frames*, however, is a genuine inequality (AM–GM–HM);

the comparison is the non-invariant content. The same mechanism produces Rényi's entropies, which Rényi obtained precisely as quasi-arithmetic averages of probabilities [3].

5 New tools from non-invariants

5.1 Concentration: φ-Chernoff bounds

Let φ be increasing and non-negative. Since $\{X \geq a\} = \{\varphi(X) \geq \varphi(a)\}$, Markov's inequality gives the *φ-concentration bound*

$$\Pr(X \geq a) \;\leq\; \frac{\mathbb{E}[\varphi(X)]}{\varphi(a)}. \tag{1}$$

Three ingredients, cleanly separated by the framework, make this powerful: (i) for sums of independent variables the *factorization* $\mathrm{MGF}_{X+Y} = \mathrm{MGF}_X\,\mathrm{MGF}_Y$ holds — an invariant identity, unique to $\varphi = \exp$ (the $+ \to \times$ homomorphism); (ii) the *convexity* of exp — a non-invariant analytic property; (iii) *optimization* over the generator. Choosing $\varphi = e^{tx}$ and minimizing over t gives Chernoff; choosing $\varphi = x^k$ gives the order-k Markov bound. The optimal member of the family is *tail-adaptive*.

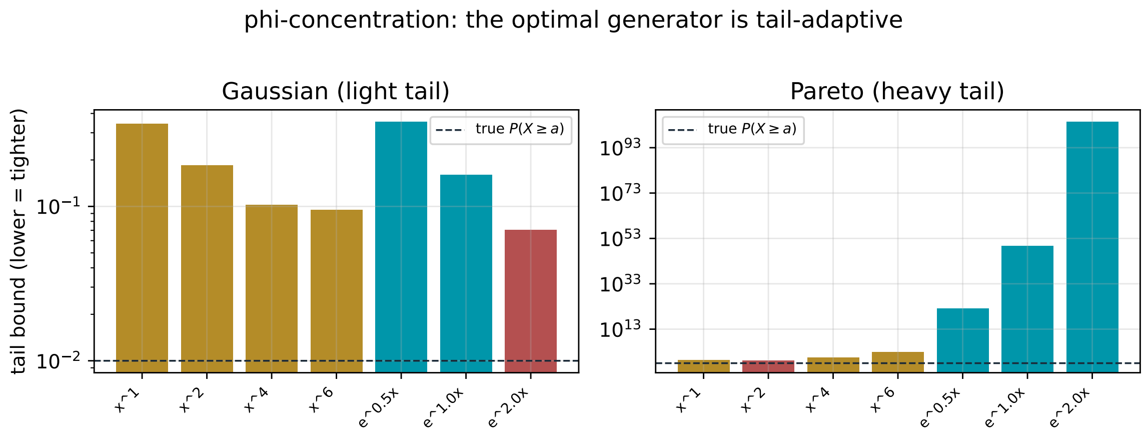

Figure 3: φ-concentration bound (1) per generator, for a light tail (Gaussian) and a heavy tail (Pareto). Exponential generators (teal) win for the light tail; polynomial generators (gold) win for the heavy tail; the tightest is highlighted (rose), the true tail dashed.

Reading. The figure is the decision of Article B carried into probability: there is no universally best generator, but the optimal one is read off the data's tail. For sub-exponential tails the exponential frame (Chernoff) is tightest; for heavy tails, where the MGF is effectively useless, a polynomial moment wins.

Remark 4 (Honest separation). The factorization (i) is invariant and is what makes the bound *tensorize* over independent data; the convexity (ii) and the optimization (iii) are where the exponential decay and the tightness come from. The framework's value is to exhibit which ingredient is which.

5.2 The multiplicative frame: Hilbert monoids and the twisted zeta

The twin product $a \odot_\varphi b = \varphi^{-1}(\varphi(a)\varphi(b))$ is, by construction, conjugate to multiplication; the twisted arithmetic is a multiplicative monoid. For an affine generator $\varphi(x) = cx + d$ we identify it exactly, with a correction.

Proposition 5 (Affine twin arithmetic is a Hilbert monoid)**.** *The unit of $\odot_\varphi$ is $\varphi^{-1}(1) = (1-d)/c$,* not 1*. With this unit, and when $d^2 \equiv d \pmod c$, the affine twin arithmetic is isomorphic via φ to the multiplicative monoid $M_{c,d} = \{m \geq 1 : m \equiv d \pmod c\}$ of an arithmetic progression. Its irreducibles are the primes of the progression; for $(c,d) = (2,1)$ these are the odd primes, so the corrected twisted primes are $\{n : 2n+1$ is prime$\}$.*

Remark 6 (A correction). The earlier numeric framework declared n twisted-prime using 1 as the unit, counting spurious primes. Using the correct unit $\varphi^{-1}(1)$ changes the $(2,1)$ spectrum from a dense set to the sparse, Sophie-Germain-flavoured $\{n : 2n+1 \text{ prime}\}$ (Figure 4, left).

The Hilbert monoids $M_{c,d}$ are the classical examples of *non-unique factorization*: in $M_{3,1} = \{1, 4, 7, 10, \dots\}$, $100 = 4 \cdot 25 = 10 \cdot 10$; in $M_{4,1} = \{1, 5, 9, \dots\}$, $441 = 9 \cdot 49 = 21 \cdot 21$ — all factors irreducible. The twisted-zeta Euler product then *overcounts* by exactly the multiplicity of factorizations: with $r(m)$ the number of factorizations of m into monoid primes,

$$\prod_{p \text{ prime in } M} (1-p^{-s})^{-1} - \sum_{m \in M} m^{-s} = \sum_{m \in M} (r(m)-1)\, m^{-s} =: \Delta(s), \tag{2}$$

the *Euler-product defect*, which vanishes iff factorization is unique (Figure 4, right).

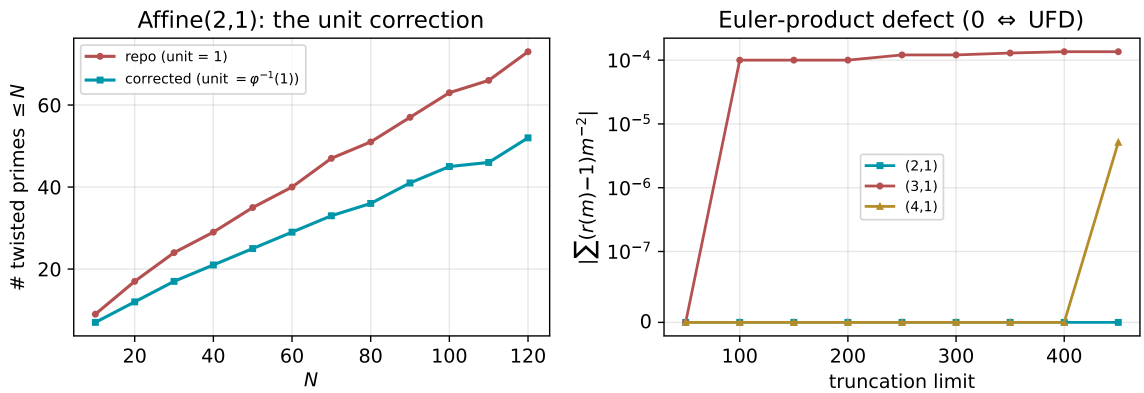

Figure 4: Left: affine $(2,1)$ twisted primes — the corrected count ($\{n : 2n+1 \text{ prime}\}$, sparse) versus the uncorrected unit-1 count (dense). Right: the Euler-product defect $|\Delta(2)|$ versus truncation; zero for $(2,1)$ (the odds, a UFD), positive and growing for the Hilbert monoids $(3,1), (4,1)$.

Reading. The left panel quantifies the unit correction; the right panel turns "unique factorization or not" into a measured number. Where $\Delta(s) = 0$ the twisted zeta is a genuine Euler product (a relabeled L-function over the progression); where $\Delta(s) \neq 0$ the multiplicative and ordering structures disagree, and the zeta is no longer a simple product.

Remark 7 (Where new zeta functions live). Within one coherent frame the zeta is a relabeled Dirichlet series over an arithmetic progression — new only by restriction. Genuine novelty requires *crossing* frames, where the additive and multiplicative structures come from different conjugations and the defect $\delta_\varphi \neq 0$: this is the level-p zeta ζ_p of the originating project. The present analysis locates that object precisely on the map of frames.

6 Discussion

The equivariance "tautology" is real but bounded: it rules within a single coherent frame, where a twin operation is a relabeling. The productive zone is its complement — the non-invariant signature (metric, measure, order, a second operation) and the comparison *across* frames. There, the framework organizes and generates genuine mathematics: the power-mean inequality from comparing frames, φ-concentration with tail-adaptive selection from the metric non-invariant, and the multiplicative theory of arithmetic progressions — with its corrected unit, its non-unique factorization, and its Euler-product defect — from the multiplicative frame. The user's intuition that "Pythagoras everywhere" and "Fourier as a special case" are major is thus correct and, here, made computational: the value is not a new abstract theorem but a systematic, measurable machine for instantiating invariants and exploiting non-invariants.

A Derivations

Proposition 1. Ψ_g is an isomorphism of the structure $(E, \odot)$; by the isomorphism transfer of model theory, every property in the language $\{\odot\}$ holds in the source iff in the image, so it is invariant. A property referring to additional symbols $(d, \mu, \le, \star)$ is evaluated in a richer structure on which Ψ_g need not be an isomorphism; it transports iff Ψ_g also preserves that symbol (an isometry for d, measure-preserving for μ, monotone for $\le$, a homomorphism for $\star$). $\square$

Proposition 3 (power-mean inequality). For $0 < p \le q$ put $y_i = x_i^p > 0$. Since $q/p \ge 1$, the map $t \mapsto t^{q/p}$ is convex, so Jensen's inequality gives $\frac{1}{n}\sum_i y_i^{q/p} \ge \left(\frac{1}{n}\sum_i y_i\right)^{q/p}$. Raising to the power $1/q$,

$$M_q = \Big(\tfrac{1}{n}\sum_i x_i^q\Big)^{1/q} = \Big(\tfrac{1}{n}\sum_i y_i^{q/p}\Big)^{1/q} \ge \Big(\tfrac{1}{n}\sum_i y_i\Big)^{1/p} = M_p.$$

The same convexity argument covers negative orders, and the endpoints are the limits $M_p \to \max_i x_i$ as $p \to +\infty$ and $M_p \to \min_i x_i$ as $p \to -\infty$ (the extreme term dominates), while $M_0 = \lim_{p\to 0} M_p = (\prod_i x_i)^{1/n}$ is the geometric mean, since $\log M_p = \frac{1}{p}\log(\frac{1}{n}\sum_i x_i^p) \to \frac{1}{n}\sum_i \log x_i$. Hence M_p is non-decreasing in p. $\square$

φ-Chernoff (1). For increasing non-negative φ, $\{X \ge a\} = \{\varphi(X) \ge \varphi(a)\}$; applying Markov to the non-negative variable $\varphi(X)$ gives $\Pr(X \ge a) = \Pr(\varphi(X) \ge \varphi(a)) \le \mathbb{E}[\varphi(X)]/\varphi(a)$. For $\varphi = e^{tx}$ and independent $X_1, \ldots, X_n$, $\mathbb{E}[e^{t\sum X_i}] = \prod_i \mathbb{E}[e^{tX_i}]$ since exp is the $+ \to \times$ homomorphism; minimizing $e^{-ta}\prod_i \mathbb{E}[e^{tX_i}]$ over $t > 0$ is Chernoff. $\square$

Proposition 5. The neutral element of $\odot_\varphi$ satisfies $\varphi(e) = 1$, i.e. $e = \varphi^{-1}(1) = (1-d)/c$. The image $\varphi(\mathbb{Z}_{\ge 1}) = \{cn + d\}$ is closed under $\times$ iff $(c\alpha + d)(c\beta + d) \equiv d \pmod{c}$, i.e. $d^2 \equiv d$; then φ is an isomorphism onto $(M_{c,d}, \times)$, and irreducibles correspond. For $(2,1)$, $M = \{\text{odds}\}$ is divisor-closed, so its irreducibles are the odd primes and the corrected twisted primes are $\{n : 2n+1 \text{ prime}\}$. $\square$

Euler-product defect (2). Expanding the product over monoid primes counts each m once per factorization, giving $\sum_m r(m)m^{-s}$; subtracting $\sum_m m^{-s}$ leaves $\sum_m (r(m)-1)m^{-s}$, which is 0 iff $r \equiv 1$, i.e. unique factorization. $\square$

B Twisted-prime density and the Dirichlet connection

The multiplicative frame is not merely a relabeling of the primes: in the saturated case it places the twisted arithmetic squarely inside analytic number theory, with an explicit zeta and an explicit prime-density law. We make both quantitative.

Density (a prime number theorem in progression). For $(c, d) = (2, 1)$ the corrected twisted primes are $\{n : 2n + 1 \text{ prime}\}$, so their counting function is *exactly*

$$\pi_\varphi(N) = \#\{n \le N : 2n + 1 \text{ prime}\} = \pi(2N + 1) - 1. \tag{3}$$

By the prime number theorem $\pi_\varphi(N) \sim \mathrm{Li}(2N+1) \sim 2N/\log(2N)$; the computed counts track $\mathrm{Li}(2N+1)$ closely and the leading term with the usual slow PNT convergence (Table 3, Figure 5 left).

Table 3: Twisted-prime density for $(2, 1)$ (computed): $\pi_\varphi(N) = \pi(2N+1)-1$ versus the leading PNT term.

N	$\pi_\varphi(N)$	$2N/\log(2N)$	ratio
10^3	302	263.1	1.148
10^4	2261	2019.5	1.120
10^5	17983	16385.3	1.097

The twisted zeta is a Hurwitz/L object. The full monoid Dirichlet series is, in closed form,

$$\zeta_{M_{c,d}}(s) = \sum_{\substack{n \ge 1 \\ n \equiv d\,(c)}} n^{-s} = c^{-s}\,\zeta(s, \tfrac{r}{c}) = \frac{1}{\varphi(c)} \sum_{\chi \bmod c} \overline{\chi}(d)\,L(s, \chi), \tag{4}$$

with $r \in \{1, \dots, c\}$ the residue of d and $\zeta(s, \cdot)$ the Hurwitz zeta; for $(2, 1)$ this is $(1-2^{-s})\zeta(s)$ and for $(3, 1)$ it is $3^{-s}\zeta(s, \frac{1}{3})$. The partial sums converge to (4) (Table 4, Figure 5 right), confirming numerically that the twisted zeta of the multiplicative frame is an explicit combination of Dirichlet L-functions.

Table 4: Twisted zeta versus its closed Hurwitz form (computed, 2–3 $\times 10^5$ terms).

(c, d)	s	partial sum	closed form $c^{-s}\zeta(s, r/c)$	error
$(2, 1)$	2	1.2336981	1.2337006	2.5×10^{-6}
$(2, 1)$	3	1.0517998	1.0517998	6×10^{-12}
$(3, 1)$	2	1.1217319	1.1217330	1.1×10^{-6}
$(3, 1)$	3	1.0207800	1.0207800	2×10^{-12}

Reading. The left panel says the twisted primes are not exotic: they are the primes of an arithmetic progression, governed by Dirichlet's theorem and the PNT. The right panel says the twisted zeta is, in the saturated frame, a known L-function combination. The novelty is therefore *not* here but at the boundary: when the residue class fails to be divisor-closed (the Hilbert monoids $M_{3,1}, M_{4,1}$), the monoid becomes a Beurling arithmetical semigroup, factorization is non-unique, and the Euler-product defect $\Delta(s)$ of (2) measures the deviation of the twisted zeta from a Dirichlet L-product — the precise analytic signature of a genuinely new arithmetic.

C Reproducibility

The analytic and multiplicative layers (`arithmetics.symbolic.twin. analysis`, `.multiplicative`) implement every object; the figures are produced by `papers/make_figures_C.py` from computed data, and each result is paired with a unit test. The whole repository passes 226 tests.

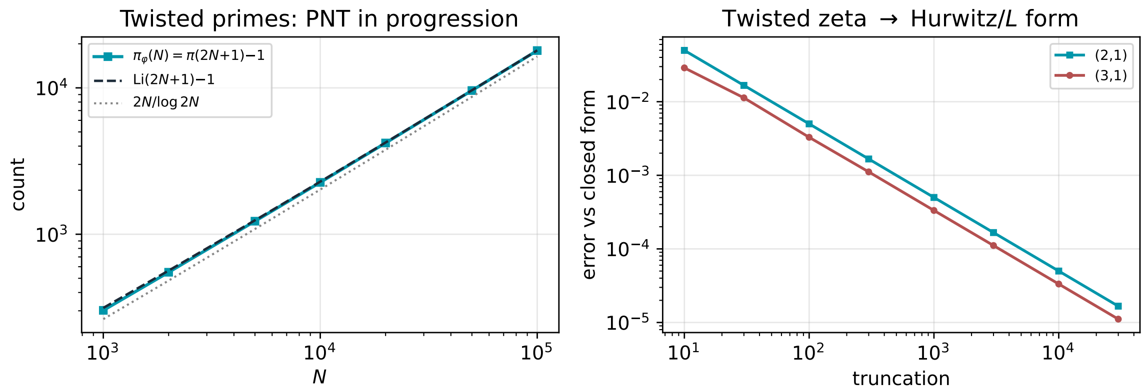

Figure 5: Left: the twisted-prime counting function $\pi_\varphi(N) = \pi(2N+1)-1$ against $\mathrm{Li}(2N+1)$ and the leading term $2N/\log 2N$. Right: the partial twisted zeta converging to its closed Hurwitz/L form for $(2,1)$ and $(3,1)$.

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